



Figure 1: NoSQL databases

CouchDB is popularly known as 'relax DB'. The design principle behind CouchDB is that it allows developers to relax by providing so many built-in features. As you start Apache CouchDB, you will be presented with the message, "Apache CouchDB started. It's time to relax."

CouchDB is well suited for applications that handle self-contained data. The traditional RDBMS stores the related data values scattered over various places, which are linked by keys. For example, in a product bill, the product details might be stored in a separate table, which is linked by the product ID field. Similarly, the vendor details might go into a separate table, which is linked by the vendor ID field. All this data is pulled by firing SQL queries, which join the related values and present them to the user. Though this application has lots of advantages such as redundancy elimination, integrity, etc, it is not needed in all places.

In CouchDB, the data is stored as a self-contained document. This self-containment mimics real world documents such as bills, tickets, business cards, etc. For example, if you take a movie ticket document, all the relevant fields are present in the ticket itself, so it makes the user understand the semantics. It doesn't require the user to link some other external data. These types of documents are called self-contained.

CouchDB attempts to mimic this real world nature by building the data store as a collection of self-contained documents represented by JSON (JavaScript Object Notation).

CouchDB installation

The recent stable version, CouchDB 1.6.1, can be downloaded from the home page of the official CouchDB website <http://couchdb.apache.org/>. CouchDB can be installed in all major operating systems such as Linux, Windows or Mac. Being an open source application, the code is also available.

The dependencies for CouchDB installation are listed below:

- Erlang OTP (>=R14B01, <R17)
- ICU
- OpenSSL
- Mozilla SpiderMonkey (1.8.5)
- GNU Make
- GNU Compiler Collection
- libcurl
- help2man
- Python (>=2.7) for docs
- Python Sphinx (>=1.1.3)

These dependencies can be installed with the following commands:

```
sudo apt-get install build-essential
sudo apt-get install erlang-base-hipe
sudo apt-get install erlang-dev
sudo apt-get install erlang-manpages
sudo apt-get install erlang-eunit
sudo apt-get install erlang-nox
sudo apt-get install libicu-dev
sudo apt-get install libmozjs-dev
```

```
sudo apt-get install libcurl4-openssl-dev
```

After the installation of the dependencies, execute the following code:

```
./configure
make&&sudo make install
```

If there are no issues, then you will be greeted with the following message:

"You have installed Apache CouchDB; it's time to relax."

To run CouchDB, fire the following command:

```
sudo -i -u couchdbcouchdb
```

In a Windows PC, you can download the .exe file and run the installer.

The CouchDB interface

CouchDB can be accessed either by using the terminal commands or the Web based interface. Both these approaches are explored in this section. To access it from the terminal, use *curl* and *http*, as shown below:

```
curlhttp://127.0.0.1:5984/
This will give you the following response:
{
  "couchdb":"Welcome",
  "uuid":"d9b0cf5a3deee84afadb52f9a30664a2",
  "version":"1.6.1",
  "vendor":{"version":"1.6.1-1", "name":"Homebrew"}
}
```

To list all the databases, enter the following commands:

```
curl -X GET http://127.0.0.1:5984/_all_dbs
[{"_replicator","_users","baseball","mycontacts","test_suite_db","test_suite_db2"}]
```

To create a new database named 'osfy', execute the following command: